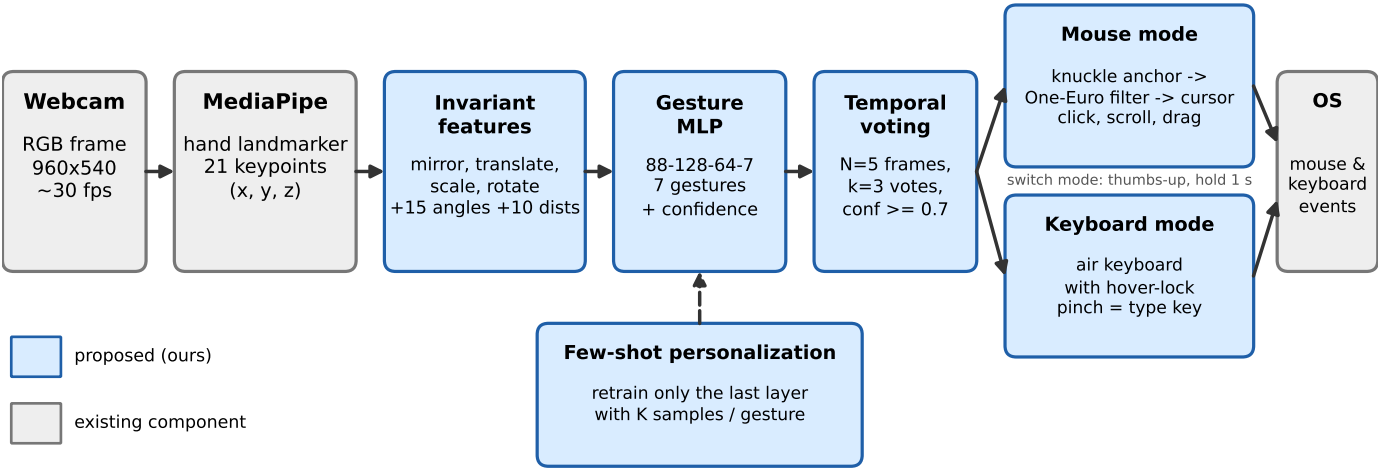
Webcam
RGB frame
960x540
~30 fps
MediaPipe
hand landmarker
21 keypoints
(x, y, z)
Invariant features
mirror, translate,
scale, rotate
+15 angles +10 dists
Gesture MLP
88-128-64-7
7 gestures
+ confidence
Temporal voting
N=5 frames,
k=3 votes,
conf >= 0.7
Mouse mode
knuckle anchor ->
One-Euro filter -> cursor
click, scroll, drag
switch mode: thumbs-up, hold 1 s
Keyboard mode
air keyboard
with hover-lock
pinch = type key
OS
mouse &
keyboard
events
Few-shot personalization
retrain only the last layer
with K samples / gesture
proposed (ours)
existing component